

CPU Scheduling Simulator and Analysis

Course: Operating Systems (Project 1)

Date: March 24, 2026

1. Algorithms and Tie-Breaking Rules

This simulator implements four fundamental CPU scheduling algorithms. To ensure deterministic results and handle edge cases where multiple processes are eligible, the following rules are applied:

Algorithms Implemented:

- **FCFS (First-Come First-Served):** A non-preemptive algorithm where the CPU is allocated to processes in the order of their arrival.
- **SJF (Shortest Job First) - Non-Preemptive:** Selects the process with the smallest burst time from the ready queue.
- **Priority Scheduling - Non-Preemptive:** Selects the process with the highest priority (represented by the lowest numerical value).
- **Round Robin (RR) - Preemptive:** Processes are assigned a fixed time quantum ($q=2$). If a process does not finish within this slice, it is moved to the back of the queue.

Tie-Breaking Rules:

- **SJF:** Smallest Burst Time → Smallest Arrival Time → Smallest PID.
- **Priority:** Lowest Priority Value → Smallest Arrival Time → Smallest PID.
- **Idle Periods:** If no process is available in the ready queue, the CPU remains IDLE until the next process arrives.
- **Ready Queue (RR):** Uses a FIFO queue where newly-arrived processes are appended as they arrive.

2. Outputs for Dataset A

Gantt Charts:

FCFS: 0-7 P1 | 7-11 P2 | 11-12 P3 | 12-16 P4 | 16-19 P5

SJF: 0-7 P1 | 7-8 P3 | 8-11 P5 | 11-15 P2 | 15-19 P4

Priority: 0-7 P1 | 7-11 P2 | 11-14 P5 | 14-18 P4 | 18-19 P3

RR (q=2): 0-2 P1 | 2-4 P2 | 4-6 P1 | 6-7 P3 | 7-9 P2 | 9-11 P4 | 11-13 P5 | 13-15 P1
| 15-17 P4 | 17-18 P5 | 18-19 P1

Average Metrics (Dataset A):

Algorithm	Avg Wait Time (WT)	Avg Turnaround Time (TAT)	Avg Response Time (RT)
FCFS	5.80	9.60	5.80
SJF	4.80	8.60	4.80
Priority	6.60	10.40	6.60
RR (q=2)	6.80	10.60	2.20

3. Outputs for Dataset B

Gantt Charts:

FCFS: 0-20 P1 | 20-22 P2 | 22-23 P3 | 23-26 P4 | 26-28 P5 | 28-29 P6

SJF: 0-20 P1 | 20-21 P3 | 21-22 P6 | 22-24 P2 | 24-26 P5 | 26-29 P4

Priority: 0-20 P1 | 20-22 P2 | 22-25 P4 | 25-26 P6 | 26-27 P3 | 27-29 P5

RR (q=2): 0-2 P1 | 2-4 P2 | 4-5 P3 | 5-7 P1 | 7-9 P4 | 9-11 P5 | 11-12 P6 | 12-14 P1
| 14-15 P4 | ... | 27-29 P1

Average Metrics (Dataset B):

Algorithm	Avg Wait Time (WT)	Avg Turnaround Time (TAT)	Avg Response Time (RT)
FCFS	17.17	22.00	17.17
SJF	16.17	21.00	16.17
Priority	17.33	22.17	17.33

Algorithm	Avg Wait Time (WT)	Avg Turnaround Time (TAT)	Avg Response Time (RT)
RR (q=2)	5.17	10.00	2.83

4. Comparison and Reflection

The analysis of both datasets reveals distinct performance trade-offs between the scheduling policies. In **Dataset A** (mixed arrivals), **SJF proved the most efficient** by minimizing average waiting time (4.80) and turnaround time (8.60). However, **Round Robin** provided the best responsiveness, with an average response time of 2.20.

The **Dataset B** stress test highlights the **convoy effect** in non-preemptive algorithms. Because P1 has a long burst time (20), it forces all subsequent short jobs to wait, resulting in high average wait times (approx. 16-17). **Round Robin (q=2)** effectively mitigated this by preempting P1, allowing short jobs to finish significantly earlier and reducing the average wait time to **5.17**. This demonstrates that while SJF is optimal for throughput, Round Robin is essential for fairness and interactive responsiveness in systems with high-variance workloads.